

10-Mark Questions & Answers — DBMS Module 4: Relational Database Design

Q1. Explain normalization from 1NF to BCNF in detail with a step-by-step unnormalized enterprise example.

Step 1: Unnormalized Relation (UNF)

`EMP_PROJ(Emp_ID, Emp_Name, Phones, Proj_ID, Proj_Name, Hours, Dept_ID, Dept_Name)`

Step 2: Convert to 1NF (Atomic Cells)

Flatten comma-separated `Phones` and `Proj_ID` entries into individual atomic rows. Primary Key = `{Emp_ID, Proj_ID}`.

Step 3: Convert to 2NF (Remove Partial Dependencies)

- Partial FDs: `Emp_ID -> Emp_Name, Dept_ID, Dept_Name` and `Proj_ID -> Proj_Name`.
- Decompose into:
 - `EMPLOYEE(Emp_ID, Emp_Name, Dept_ID, Dept_Name)`
 - `PROJECT(Proj_ID, Proj_Name)`
 - `ASSIGNMENT(Emp_ID, Proj_ID, Hours)`

Step 4: Convert to 3NF (Remove Transitive Dependencies)

- Transitive FD in EMPLOYEE: `Dept_ID -> Dept_Name`.
- Decompose EMPLOYEE into:
 - `EMP(Emp_ID, Emp_Name, Dept_ID)`
 - `DEPARTMENT(Dept_ID, Dept_Name)`

Step 5: Convert to BCNF (Strict Super Key Determinants)

- Check all determinants $X \rightarrow Y$; ensure every X is a super key.